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## Unpacking the Different Flavors of Curiosity

Disciplines as diverse as psychology, education, and neuroscience strive to understand curiosity and harness its power for human thriving. Despite interest and decades of research, however, it is noteworthy that even a consensus on the definition of curiosity remains elusive. Critically for scientists, however, the absence of a definition or taxonomy of curiosity means we lack the power to predict when curiosity occurs, and how it might alter behavior. Because of this, there is a challenge for understanding the functions of curiosity within and across species.

We propose that to fully characterize the dimensions of curiosity, we must understand how external elicitors of curiosity interact with an agent's internal state and goals, to produce not only varied information-seeking behavior but also distinct learning episodes. Building on Berlyne's initial characterizations of curiosity (Berlyne, 1954), we argue that curiosity is comprised of multiple dimensions; the interaction amongst these dimensions influences how curiosity is elicited, its associated subjective experience, and the learning behaviors and plasticity it produces. An understanding of how curiosity is induced, what creates a drive to seek information that is not directly instrumental, provides a useful starting point.

Previous work has suggested that curiosity arises from a perceived gap in one's knowledge, referred to as the 'information gap' (Loewenstein, 1994). However, this theory makes no prediction about how the size of the information gap impacts induction or the subsequent search for information. Information gaps can vary widely in size, ranging from the considerable gap associated with novelty, where stimuli have not previously been encountered and the space to acquire information is quite large, to gaps so small that a single cue can relieve tension induced by the question at hand (Blanchard, Hayden, & Bromberg-Martin, 2015). These varying sizes of gaps in knowledge imply varied behaviors for an agent navigating the potential search space.

In addition to the quantitative amount of missing information, another variable to be considered is how the state of missing information is perceived by an agent—that is, how it maps onto the agent's internal representation of the world and its affective or neuromodulatory state. According to this view, the awareness of an information gap might give rise to multiple—and possibly conflicting—motivational drives and behavioral goals. For example, we have previously demonstrated that the same external information gap gives rise to distinct brain and behavioral responses according to the valence of motivational states (Murty, LaBar, & Adcock 2016; Chiew et al., 2018).

This principle can be refined to suggest specific predictions related to motivational states, as follows: if an agent's goal in acquiring new information is to reduce environmental or decision uncertainty for an action imperative, curiosity might be precisely targeted, functioning to limit actions to only those serving to obtain goal-relevant information (Blanchard et al., 2015). Conversely, if the goal is to interrogate a novel space, resulting actions may be more exploratory, or diffuse and varied (Baranes et al., 2014). We have argued elsewhere for distinguishing these motivational states (see Murty & Adcock 2017; Dickerson & Adcock in press). Would both of these example behaviors be expected to drive behavioral specialization? Or only the first?

Understanding curiosity from a function-first perspective requires that we characterize the complex taxonomies, at multiple levels of analysis, that shape both the induction of curiosity and behavioral pursuit of information. By characterizing different stimuli, search processes, and learning episodes associated with curiosity, we may start to understand its functions in our adaptive exploration, knowledge acquisition, and skill development. Only after we delineate these multiple different flavors of curiosity can we begin to assess whether and how curiosity may drive extreme specialization, in humans and other animals.

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